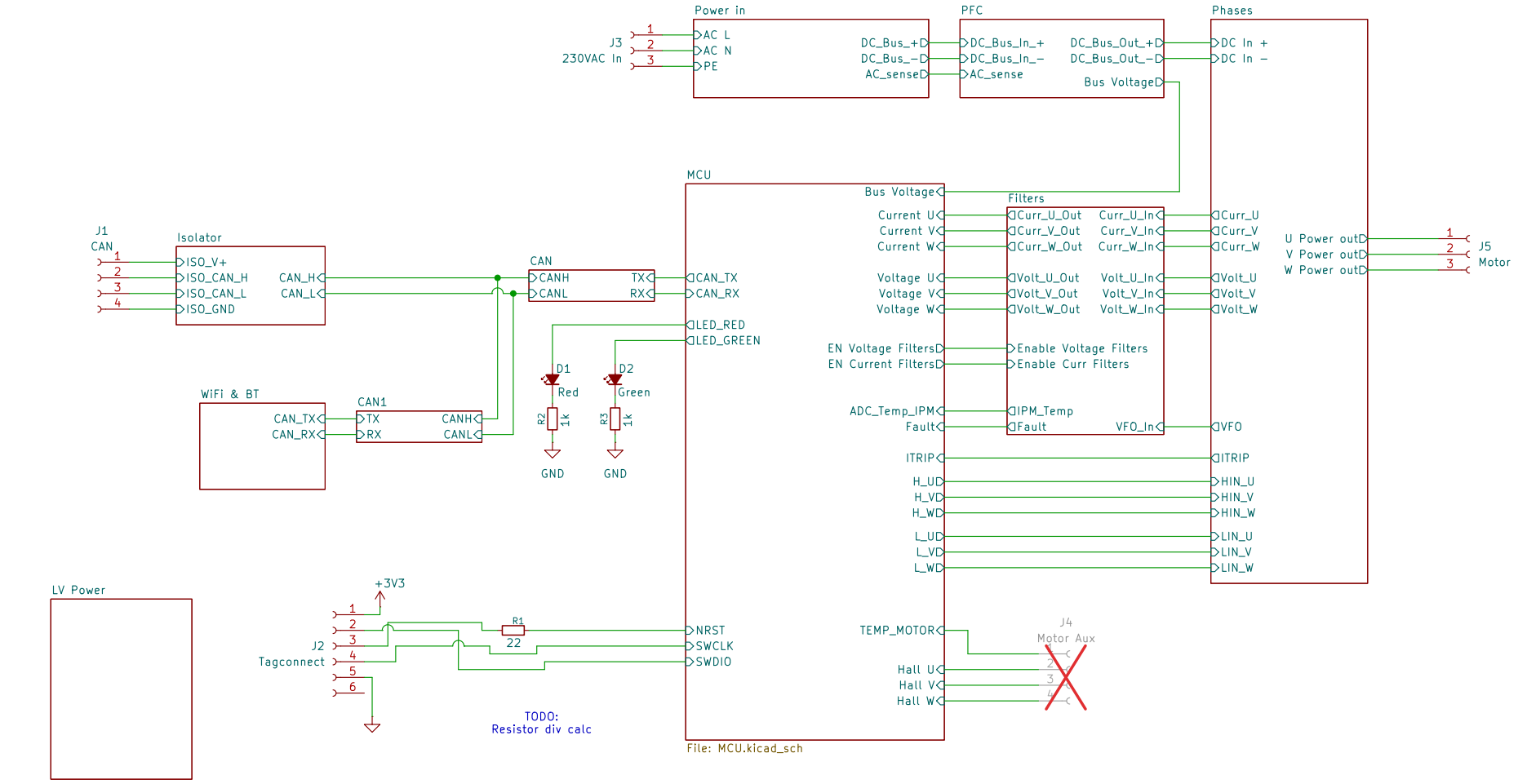


H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

HVAC VESC

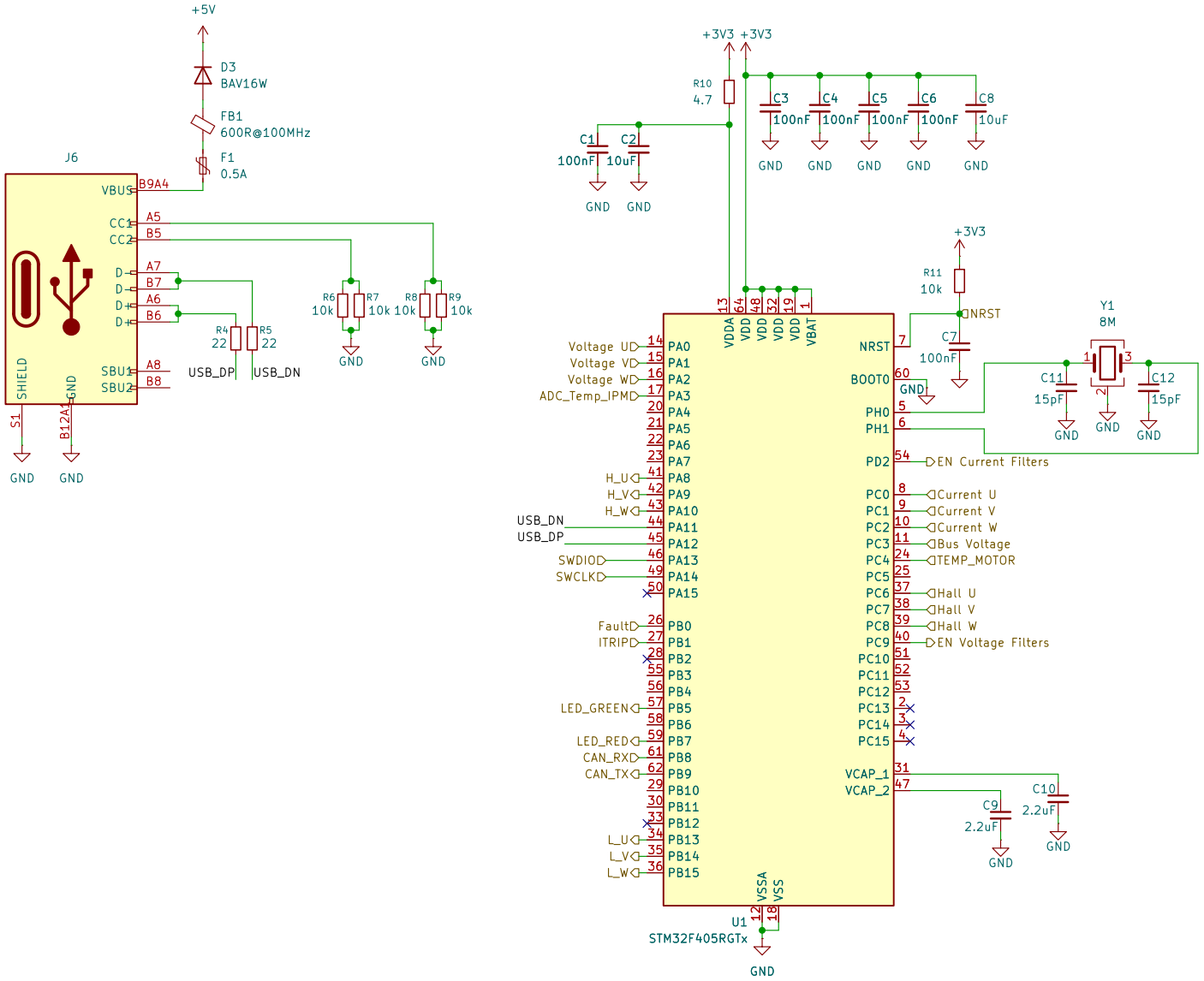
Needs to follow for CE cert:
EN IEC 63000 (RoHS)
IEC 60335-2-34 (Particular requirements for motor-compressors)
IEC_61000-3-2 (Limits for harmonic current emissions) <--- Most important

Picobello B.V.

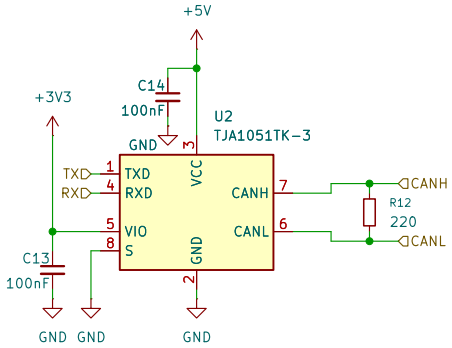
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File: CompressorController.kicad_sch

Title: Mains Powered VESC MD

Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.4		Id: 1/11



Picobello B.V.		
Sheet: /MCU/		
File: MCU.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.4		Id: 2/11



Picobello B.V.		
Sheet: /CAN1/ File: can.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.4		Id: 3/11

DC In +

DC In −

HV

DC Bus +

DC Bus −

+15V

C18 2.2uF

C19 100nF

C23 1nF

GND

U6

IKCM10H60GAXKMA1

VDD

VSS

NC

VS(U)

VB(U)

VS(V)

VB(V)

VS(W)

VB(W)

HIN(U)

HIN(V)

HIN(W)

LIN(U)

LIN(V)

LIN(W)

VFO

ITRIP

P

U

V

W

NU

NV

NW

C20 10uF

C24 100nF

C21 10uF

C25 100nF

C22 10uF

C26 100nF

C28 470nF

HIN_UD R13 ZZ

HIN_VD R14 ZZ

HIN_WD R15 ZZ

LIN_UD R16 ZZ

LIN_VD R17 ZZ

LIN_WD R18 ZZ

R19 R20 R21 R22 R23 R24 10k

VFO

ITRIP

R25 1k

C27 1nF

IKCM10H60GA

C3615235

U Power out

V Power out

W Power out

Volt_U

Volt_V

Volt_W

R29 R30 R31 200k

R32 3.16k

R33 R34 R35 200k

R36 3.16k

R37 R38 R39 200k

R40 3.16k

+3V3

C15 100nF

TP7

Curr_U

U3

INA240A1D

R97 50mR

R103 50mR

R104 50mR

C16 100nF

TP8

Curr_V

U4

INA240A1D

C17 100nF

TP9

Curr_W

U5

INA240A1D

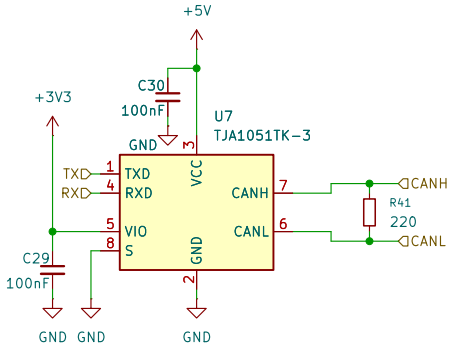
Picobello B.V.

Sheet: /Phases/

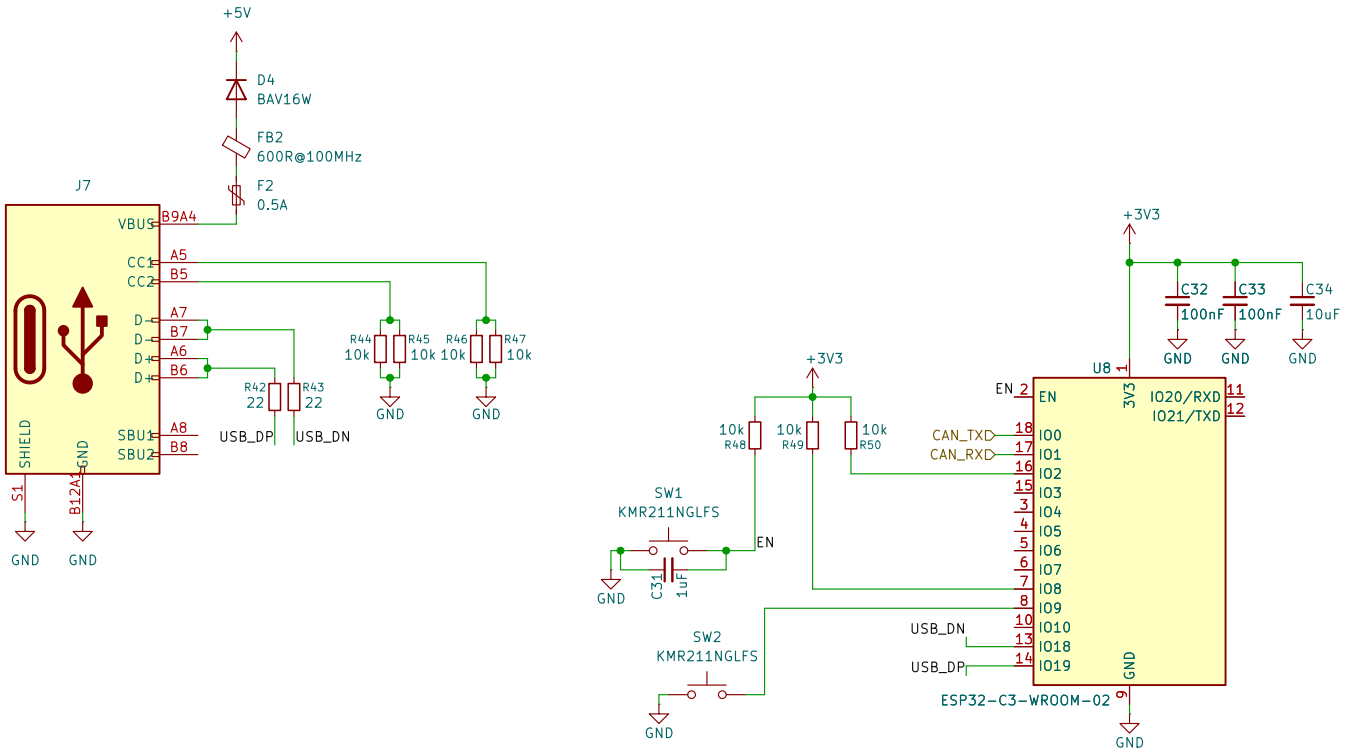
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Title: Mains Powered VESC MD

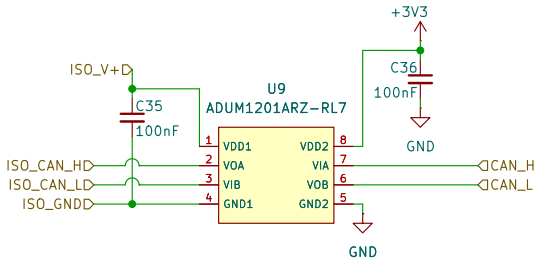
Size: A3	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.4		Id: 4/11



Picobello B.V.		
Sheet: /CAN/		
File: can.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.4		Id: 5/11

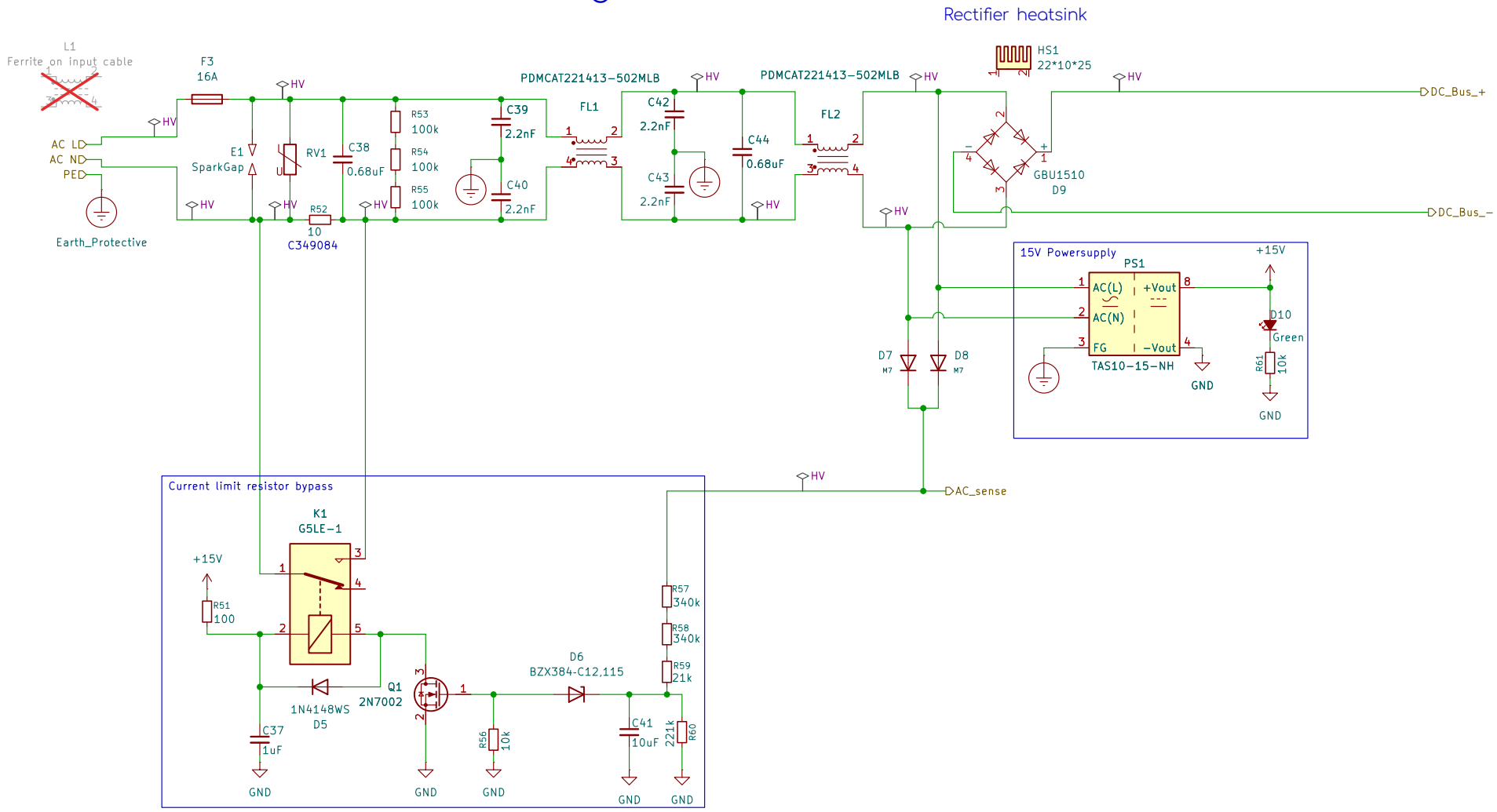


Picobello B.V.		
Sheet: /WiFi & BT/		
File: express.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.4		Id: 6/11



Picobello B.V.		
Sheet: /Isolator/ File: isolator.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.4		Id: 7/11

Following TIDA-00443



Picobello B.V.		
Sheet: /Power in/		
File: powerin.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.4		Id: 8/11

Following TIDA-00443

Heatsink for D14

D12
1N5406

HS2
XSD683-022

D14
DSEP30-06A

R77
43

C69
330pF

DC_Bus_In_+

DC_Bus_In_-

DC_Bus_Out_+

DC_Bus_Out_-

HV

L2
330uH

C51
1.5uF

R69
0.2

R68
220

R70
0.02

IDk why, but TI ref design does this, trim?

Gate drive

U16
UCC27517ADBV

+15V

C55
10uF

C56
100nF

GND

D13
MMSD914

D18
MMSD914

D19
MMSD914

Q3 IPL60R185C7

Q4 IPL60R185C7

Q5 IPL60R185C7

R74
2

R75
1

R98
2

R99
1

R101
2

R102
1

R76
10k

C72
220uF

C73
220uF

C75
220uF

Bus Voltage

R84
200k

R85
200k

R86
200k

R87
3.16k

R92
221k

R94
221k

Bleeder boys

R93
221k

R95
221k

TI's Fet got quite hot (90C)
So 3x parallel, and better FETs as well

PFC controller

U15
UCC28180D

GND
ICOMP
ISENSE
FREQ
GATE
VCC
VSENSE
VCOMP

+15V

C54
2.7nF

R71
21.5k

R73
23.7k

C61
4.7uF

C62
390nF

C64
100nF

C65
10uF

Vsense

TP5
HVDC-Sense

R88
332k

R89
332k

R90
340k

R91
13k

C76
820pF

Boost follower control and sensing

AC_sense

R62
332k

R63
332k

R64
340k

TP6
ACSense

AC Sense

R65
10k

C46
10uF

D11
BAT54J

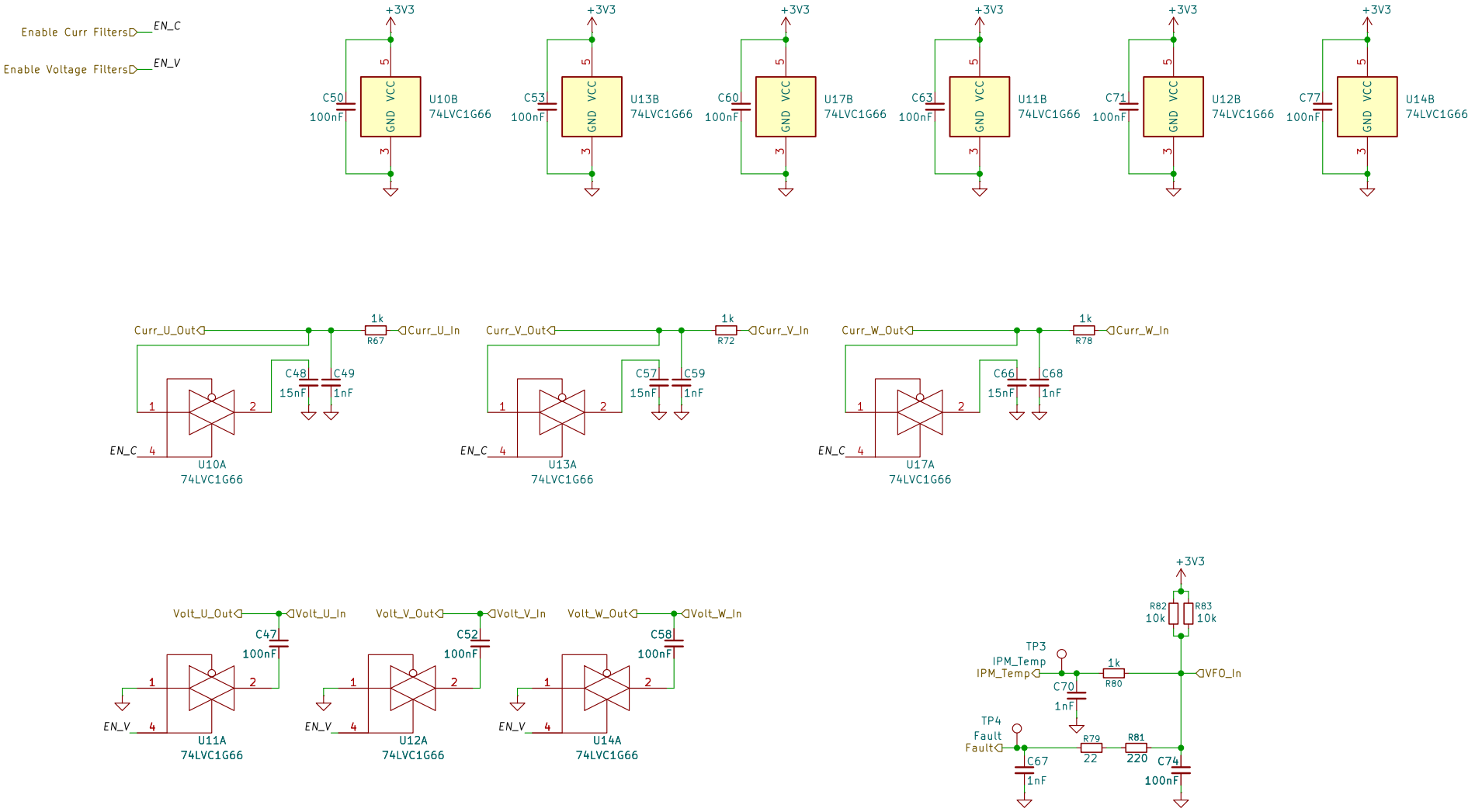
Q2
BC847C

R66
13.7k

Vsense

Boost follower is actually quite a bit more efficient.
But the software might not like the regulation loop.

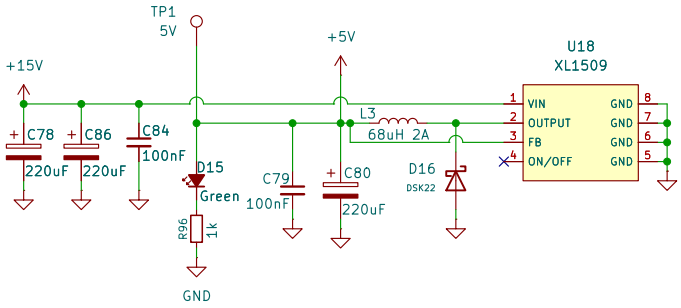
Picobello B.V.	
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File: pfc.kicad_sch	
Title: Mains Powered VESC MD	
Size: A3	Date: 2026-04-16
KiCad E.D.A. 10.0.4	Rev: R1
	Id: 9/11

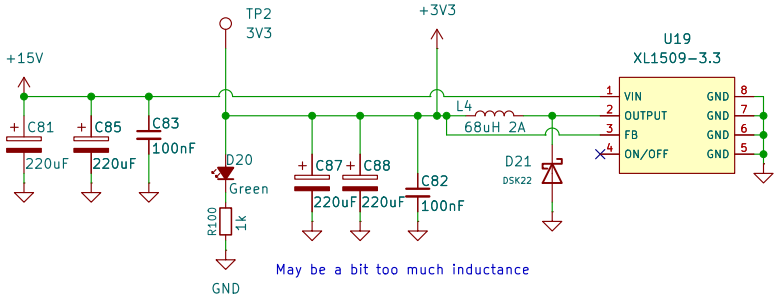


VFO Pin is also UVLO and OC fault output

Picobello B.V.	
Sheet: /Filters/	
File: filters.kicad_sch	
Title: Mains Powered VESC MD	
Size: A4	Date: 2026-04-16
KiCad E.D.A. 10.0.4	Id: 9/11

Rev: R1





Picobello B.V.		
Sheet: /LV Power/ File: lvpower.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.4		Id: 10/11